4.5 Scatter Plots and Lines of Fit

Name _____ Period ____ Date _____

Algebra I 4.5 (30-minute plan) · Student sheet

I can…

- interpret scatter plots and identify correlation.
- draw and write an equation for a line of fit.
- use a line of fit to make predictions.

Vocabulary · fill in the missing word

Scatter plot — a graph of _____ data as points

Positive / negative correlation — y tends to increase / decrease as x _____

No correlation — no clear _____

Line of fit — a line that _____ the trend of the data

Part 1 · Notes and examples (fill in each note, then work the example)

Positive correlation: Positive correlation: as x increases, y tends to _____.

Example 1

Tell whether each shows positive, negative, or no correlation.

a. age of a car and its value

b. height and shoe size

c. shoe size and test score

Line of fit: A line of fit is close to _____ of the points.

Careful: Correlation does not mean _____.

Example 2

Use the study-hours data. A line of fit passes through (1, 62) and (6, 88). Write its equation.

Study hours

Hours	1	2	3	4	5	6
Score	62	68	71	79	83	88

Part 2 · Partner practice

Partner practice

Does the number of hours spent watching TV and the number of hours sleeping likely show positive, negative, or no correlation?